SCHEDULED PLAN OF ACTIVITIES

Visiting Expert Program (HRDP – PCIEERD)

Visiting Expert: Dr. Jeffrey S. Perez
Affiliation: Philippine Institute of Volcanology and Seismology (DOST–PHIVOLCS)
Project Title: *Training and Development in Data-Driven Mathematical Modeling for Earthquake Epicenter Estimation in Pre-Instrumental and Contemporary Seismic Events*
Duration Covered: May – September 2026

A. SCHEDULED PLAN OF ACTIVITIES

Month	Activity	Description of Activities	Expected Outputs / Deliverables	Responsible Unit / Person
May 2026	Foundations of Earthquake Processes, Observations, and Hazard Interpretation	Conduct introductory interdisciplinary sessions discussing the physical foundations of earthquakes, seismic waves, earthquake observations, intensity interpretation, and hazard-related concepts. Discussions will emphasize the distinction between earthquake source mechanisms, wave propagation, observational data, and damage interpretation. Participants will be introduced to earthquakes as measurable, uncertain phenomena relevant to mathematical, computational, and geoscientific analysis.	Compiled Learning Portfolio; Session Documentation; Participant Reflection Outputs; Foundational Concept Notes	Visiting Expert; Project Leader; Mathematics Department; Project Team; Participating Trainees
June 2026	Tectonic Sources and Spatial Interpretation of Philippine Earthquakes	Conduct guided discussions on tectonic structures and earthquake-generating systems in the Philippine setting. Activities include conceptual interpretation of earthquake locations, spatial variability of seismic intensity, attenuation behavior, and geological controls affecting observations. Emphasis will be placed on how spatial reasoning and observational interpretation contribute to earthquake analysis workflows.	Compiled Learning Portfolio; Spatial Interpretation Exercises; Technical Discussion Outputs; Attendance and Documentation Reports	Visiting Expert; Project Leader; Mathematics Department; Project Team; Participating Trainees
July 2026	Historical Earthquake Catalogs, Macroseismic Data, and Observational Uncertainty	Facilitate sessions examining historical earthquake records, macroseismic observations, and uncertainty in earthquake catalogs. Discussions will focus on limitations of historical data, reporting bias, incomplete observations, and interpretative challenges in reconstructing earthquake events from non-instrumental records. Participants will examine the role of uncertainty in scientific inference and historical seismic interpretation.	Compiled Learning Portfolio; Historical Data Interpretation Exercises; Reflection and Analysis Outputs; Session Documentation	Visiting Expert; Project Leader; Mathematics Department; Project Team; Participating Trainees

August 2026	Case Studies of Significant Philippine Earthquakes: Interpretation, Uncertainty, and Data-Driven Analysis	Conduct interdisciplinary case study discussions on significant Philippine earthquakes relevant to the expertise of the Visiting Expert. Sessions will analyze observational patterns, seismic intensity distributions, interpretation uncertainty, and implications for data-driven analysis and hazard understanding. Participants will examine how real earthquake events are studied through scientific and computational workflows.	Compiled Learning Portfolio; Case Study Analysis Outputs; Technical Notes and Discussions; Activity Documentation	Visiting Expert; Project Leader; Mathematics Department; Project Team; Participating Trainees
September 2026	Modern Seismological Workflows, Data Systems, and Challenges in the Philippine Context	Conduct discussions on modern seismological workflows, earthquake data systems, mapping platforms, computational tools, and scientific data management approaches relevant to seismic analysis. Emphasis will be placed on challenges in the Philippine setting, including uneven instrumentation, historical data gaps, observational uncertainty, and interdisciplinary opportunities for mathematical, computational, and data-driven system development. Participants will also be introduced to conceptual considerations relevant to the development of scientific software systems for earthquake analysis.	Compiled Learning Portfolio; Conceptual System Design Outputs; Technical Reflection Papers; Session Documentation and Summary Report	Visiting Expert; Project Leader; Mathematics Department; Project Team; Participating Trainees

B. DETAILED CALENDAR TIMELINE (FOR REFERENCE / INTERNAL USE)

Visiting Expert Program – 36 Hours (Saturdays Only)

Duration: May – September 2026

Format: Online (Gmeet) except final session (Face-to-face)

PHASE I – Foundations of Earthquake Processes, Observations, and Hazard Interpretation (6 HOURS)

Date	Session	Mode	Duration	Focus	Key Activities	To accomplish	Tangible Output
May 23, 2026	1	Online	2 hours	Physical Foundations of Earthquakes and Seismic Signals	Expert discussion on seismic waves, earthquake source mechanisms, wave propagation, intensity observations, physical damage, and observational uncertainty. Guided interdisciplinary discussion on earthquakes as measurable and inferential scientific phenomena.	Establish a foundational understanding of earthquake processes, observational systems, and uncertainty in seismic interpretation.	Compiled Learning Portfolio; Reflection Notes; Session Documentation
May 30, 2026	2	Online	2 hours	Earthquake Observations, Intensity, and Spatial Interpretation	Guided discussion on macroseismic observations, intensity variation, attenuation effects, local geology, and spatial interpretation of earthquake observations. Conceptual introduction to observational inference and epicenter estimation.	Understand how earthquake observations vary spatially and how seismic characteristics are inferred from incomplete and uncertain datasets.	Compiled Learning Portfolio; Spatial Interpretation Exercises; Technical Reflection Outputs
June 6, 2026	3	Online	2 hours	Earthquake Hazards, Uncertainty, and Interdisciplinary Interpretation	Interdisciplinary discussion on uncertainty in earthquake analysis, hazard interpretation, observational limitations, and scientific reasoning under incomplete information. Introduction to the role of data-driven scientific systems in earthquake interpretation.	Develop awareness of uncertainty-aware reasoning and interdisciplinary approaches in seismic interpretation and hazard-related analysis.	Compiled Learning Portfolio; Conceptual Analysis Outputs; Session Documentation and Reflection Notes

PHASE II – Tectonic Sources and Spatial Interpretation of Philippine Earthquakes (6 HOURS)

Date	Session	Mode	Duration	Focus	Key Activities	To accomplish	Tangible Output
June 13, 2026	4	Online	2 hours	Tectonic Frameworks and Earthquake-Generating Structures in the Philippines	Expert discussion on Philippine tectonic systems, subduction zones, trenches, active faults, and regional plate interactions. Guided interdisciplinary interpretation of how tectonic structures influence earthquake occurrence and seismic hazards.	Develop foundational understanding of tectonic environments and their influence on observable earthquake behavior and seismic interpretation.	Compiled Learning Portfolio; Tectonic Interpretation Notes; Session Documentation
June 20, 2026	5	Online	2 hours	Spatial Distribution of Earthquake Observations and Intensity Variability	Guided discussion on macroseismic observations, intensity variation, attenuation effects, local geology, and spatial interpretation of earthquake observations. Conceptual introduction to observational inference and epicenter estimation.	Understand how earthquake observations vary geographically and recognize the challenges associated with interpreting incomplete and spatially variable seismic information.	Compiled Learning Portfolio; Spatial Interpretation Exercises; Technical Reflection Outputs
June 27, 2026	6	Online	2 hours	Modern Seismological Workflows, Philippine Challenges, and Data-Driven Scientific Systems	Discussion on modern seismological workflows involving observational networks, earthquake catalogs, geospatial systems, visualization platforms, and computational tools. Introduction to Philippine-specific challenges and the role of interdisciplinary scientific systems in earthquake analysis and interpretation.	Develop awareness of modern seismic analysis workflows, observational limitations in the Philippine setting, and the importance of interdisciplinary and data-driven scientific systems.	Compiled Learning Portfolio; Conceptual System Analysis Outputs; Session Documentation and Reflection Notes

PHASE III – Historical Earthquake Catalogs, Macroseismic Data, and Observational Uncertainty (6 HOURS)

Date	Session	Mode	Duration	Focus	Key Activities	To accomplish	Tangible Output
July 11, 2026	7	Online	3 hours	Historical Earthquake Catalogs and Macroseismic Observations	Expert discussion on historical earthquake records, macroseismic observations, documentary evidence, intensity reports, and the development of earthquake catalogs in the Philippine setting. Guided interdisciplinary discussion on the interpretation and limitations of non-instrumental earthquake observations.	Develop foundational understanding of historical earthquake catalogs, macroseismic interpretation, and the challenges associated with incomplete and historically constrained seismic records.	Compiled Learning Portfolio; Historical Interpretation Notes; Session Documentation
July 18, 2026	8	Online	3 hours	Observational Uncertainty, Historical Interpretation, and Data-Driven Seismic Inference	Guided discussion on uncertainty in historical earthquake analysis, reporting bias, incomplete geographic coverage, inconsistent observations, and uncertainty-aware interpretation. Conceptual introduction to interdisciplinary and data-driven approaches in historical seismic inference and epicenter estimation.	Understand the role of uncertainty and observational limitations in historical earthquake interpretation and appreciate the importance of interdisciplinary and data-driven reasoning in seismic analysis.	Compiled Learning Portfolio; Uncertainty Analysis Exercises; Technical Reflection Outputs

PHASE IV – Case Studies of Significant Philippine Earthquakes: Interpretation, Uncertainty, and Data-Driven Analysis (6 HOURS)

Date	Session	Mode	Duration	Focus	Key Activities	To accomplish	Tangible Output
August 15, 2026	9	Online	3 hours	Interpretation of Significant Philippine Earthquakes and Observational Complexity	Expert-led discussion on selected significant Philippine earthquakes, including observational patterns, macroseismic intensity distributions, tectonic context, geological effects, and spatial impact variability. Guided interdisciplinary analysis of observational complexity and interpretative limitations in real earthquake events.	Develop understanding of how significant earthquake events are interpreted through observational, geological, and spatial information under real-world conditions.	Compiled Learning Portfolio; Case Study Interpretation Notes; Session Documentation
August 22, 2026	10	Online	3 hours	Observational Uncertainty, Historical Interpretation, and Data-Driven Seismic Inference	Guided discussion on uncertainty in historical earthquake analysis, reporting bias, incomplete geographic coverage, inconsistent observations, and uncertainty-aware interpretation. Conceptual introduction to interdisciplinary and data-driven approaches in historical seismic inference and epicenter estimation.	Understand the role of uncertainty and observational limitations in historical earthquake interpretation and appreciate the importance of interdisciplinary and data-driven reasoning in seismic analysis.	Compiled Learning Portfolio; Uncertainty Analysis Exercises; Technical Reflection Outputs

PHASE V – Final Face-to-Face Evaluation (6 HOURS)

Date	Session	Mode	Duration	Focus	Key Activities	To accomplish	Tangible Output
September 12, 2026	11	Onsite	3 hours	Modern Seismological Workflows, Data Systems, and Emerging Analytical Approaches	Expert discussion on modern seismological workflows involving seismic networks, earthquake catalogs, geospatial systems, computational platforms, visualization tools, and observational data systems. Guided interdisciplinary discussion on Philippine-specific seismic challenges and exploratory applications of artificial intelligence, machine learning, and time-series analysis in earthquake-related studies.	Develop awareness of contemporary computational and analytical approaches in seismology and understand the role of interdisciplinary scientific systems in earthquake analysis and interpretation.	Compiled Learning Portfolio; Technical Reflection Notes; Session Documentation
September 12, 2026	12	Onsite	3 hours	Prototype Presentation, Scientific Integration, and Interdisciplinary Review	Participant presentation of conceptual, analytical, or prototype-based outputs developed from interdisciplinary knowledge acquired throughout the training program. Guided scientific and technical review involving discussions on feasibility, uncertainty-aware reasoning, data interpretation, visualization, and future research directions.	Consolidate interdisciplinary learning outcomes through conceptual or prototype-based presentation and demonstrate understanding of earthquake interpretation and scientific system development workflows.	Prototype or Conceptual Output Presentation; Compiled Learning Portfolio; Evaluation and Feedback Documentation; Certificate of Completion